

MAT 3150 — Homework 2

Due September 17

Problem 1 (10 points)

Prove that if x and y are arbitrary real numbers such that $x < y$, there exists at least one irrational number z satisfying $x < z < y$.

Proof.

□

Problem 2 (10 points)

Find the supremum and the infimum of the sets

$$S_1 = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} \quad \text{and} \quad S_2 = \left\{ \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}.$$

S_1

S_2

Problem 3 (10 points)

If T, S are nonempty sets so that $s \leq t$ for all $s \in S$ and $t \in T$, prove that $\sup S$ and $\inf T$ exist, and prove the inequality $\sup S \leq \inf T$.

Proof.

□

Problem 4 (10 points)

For A and B nonempty bounded subsets of real numbers, let $A + B$ be the set of all sums $a + b$, where $a \in A$ and $b \in B$. Prove that

$$\sup(A + B) = \sup A + \sup B.$$

Proof. Step 1: $\alpha + \beta$ is an upper bound for $A + B$.

Step 2: $\alpha + \beta$ is the least upper bound.

□